

KEEGAN DEMPSEY, PE

[Email](#) | [LinkedIn](#) | [Website](#)

PROFESSIONAL & RESEARCH EXPERIENCE

CARNEGIE MELLON UNIVERSITY - Pittsburgh, PA, USA (June 2025–Present)

Graduate Researcher – Resource Recovery Decision Analytics, Process Systems Engineering & Machine Learning

- Developing a context aware decision framework for waste to value systems that links regional infrastructure, feedstock availability, process performance, policy constraints, and local market demand into optimization ready models.
- Built a regional data pipeline that converts public facility, wastewater, emissions, permit, road, and pipeline records into candidate nodes, links, resource classes, capacity bounds, and infrastructure constraints.
- Developing equipment level surrogate models for gasification and related conversion pathways to estimate process performance across heterogeneous feedstocks and operating conditions.
- Applying Pyomo based mathematical programming, process synthesis, and techno economic analysis methods to compare pathway portfolios under local infrastructure and market constraints.
- Built an interactive infrastructure viewer to audit inferred regional networks before using them in optimization, communication, or capital planning.

HYSATA - Port Kembla, NSW, AUS (February 2023 - August 2023)

Balance of Plant (BOP) & Systems Engineer / Techno-economic Modeling Lead

- Built load following system models from first principles and experimental electrolyzer performance data to optimize BOP configuration for diverse renewable supply profiles.
- Developed BOP process simulation in Aspen HYSYS integrating dynamic electrolyzer degradation curves to inform equipment sizing, layout, and capital cost estimation.
- Created and managed the company's integrated techno economic model, linking process performance, equipment design, cost assumptions, and scenario analysis to quantify impacts on levelized hydrogen cost and investment decisions.
- Proposed stack-level design innovations that consolidated BOP functionalities, achieving cost & performance improvements.
- Aligned technical / commercial team workflows, bridging data and scope gaps to improve cross-functional project execution.

ARUP - Melbourne, VIC, AUS & New York, NY, USA (September 2019 - January 2023)

Chemical Engineer – Energy, Transaction Advisory & Sustainability Groups

- Developed techno-economic modeling methods, electrolyzer performance models and plant process simulations for incentivizing, sizing and costing of green hydrogen projects worldwide.
- Advised governments and companies on hydrogen and synthetic fuel deployment strategies.
- Supported technical due diligence for energy and infrastructure projects, evaluating technology readiness, process assumptions, project risks and commercial claims.
- Conducted and oversaw research to develop educational tools for internal and external stakeholders.

Mechanical Engineer – Buildings, Infrastructure, & Sustainability Groups

- Built energy models to reduce building energy loads and inform design.
- Led design and analysis of mechanical scope on critical infrastructure projects globally.

BURNS & MCDONNELL - New York, NY, USA & Houston, TX, USA (May 2015 - August 2019)

Assistant Mechanical Engineer – Aviation & Federal Group

- Led front end planning through detailed design for mechanical systems on the \$4B Delta LaGuardia Airport Terminal Reconfiguration Project, coordinating piping, HVAC, and fueling scope across a large multidisciplinary infrastructure team.
- Automated mechanical design models, improving workflow efficiency for the entire project team.

Assistant Chemical Engineer – Oil, Gas & Chemicals Group

- Facilitated front end planning through detailed design oil and gas processing plant projects.
- Created automation processes for P&ID models to include data in Heat & Material balances to generate equipment, line list and pipe sizing calculations, which has been manual previously.

EDUCATION

CARNEGIE MELLON UNIVERSITY - Pittsburgh, PA, USA (2026, expected)

Master of Science, Chemical Engineering

TEXAS TECH UNIVERSITY - Lubbock, TX, USA (2015)

Bachelor of Science, Chemical Engineering (Minor: Bioengineering, Chemistry)

SKILLS

Software: Aspen (HYSYS, PLUS, etc.), IES, COMSOL, Smart Plant P&ID, Revit, UniSim, Excel.

Programming: Python (Pyomo, IDAES, causal-learn, NumPy, scikit-learn, JAX, Calliope, Pycse), GAMS, Matlab.

SELECTED PUBLICATIONS & SPEAKING ENGAGEMENTS

- UKPACT. *eThekwini Regional Hydrogen Strategy*. eThekwini Municipality Energy Office, 2023. [Link](#).
- Dempsey, K. & Gillespie, J. " *Can Synthetic Fuels (Synfuels) Help Reduce Emissions in Australia's Rail Sector While Paving the Way to a Carbon Free Future?*". Presentation at AUSRail, 6 Dec. 2022, Brisbane, QLD. Conference Paper. [Link](#).
- Department of Climate Change, Energy, the Environment and Water, and Australian Hydrogen Council. *Water for Hydrogen: Technical Paper*. Reference: 286739-00-RPT-001. 14 Nov. 2022, [Link](#).
- Dempsey, K. *Cleaner, Alternative Fuels for Transport*. Presentation at *Our Pathway Through Net Zero \ Reduce, Restore, Remove: A Call to Action*, 30 Aug. 2022, Melbourne, VIC. Video. [Link](#).
- World Bank. *Opportunities and Barriers for the Deployment of Green Hydrogen in Chile's Markets: Small and Medium Grids: Task 3 - Isolated Grid Study*. Washington, D.C.: World Bank Group, 01 Aug. 2022. [Link](#).